

OmniStore Strategic Analytics Project

Enterprise Master Data Blueprint & Architectural Dictionary

1. Project Context & Data Architecture

This document serves as the official schema reference guide and diagnostic manual for the `raw_retail_ecom_business_dataset.csv` database. This complex corporate file simulates the operational realities of **OmniStore Global Retail Group**—a cross-border multi-channel retailer specializing in Electronics, Apparel, Home Goods, and High-Margin Beauty products.

To mirror real-world constraints, the dataset bypasses idealized sanitized structures. It intentionally integrates widespread master data friction points, extraction pipeline failures, and human manual-entry errors (comprising approx. 5%–15% degradation across transactional fields). Analysts must construct an end-to-end Extract-Transform-Load (ETL) and data cleaning protocol before conducting executive strategic reviews.

Analytical Mandate: Do not just remove rows containing data issues. Standard cleaning, numeric conversion, imputation modeling, string standardization, and structural deduplication must be executed using Python, SQL, Power Query, or specialized platform functions to secure analytical integrity.

2. Deep-Dive Analytical Use Cases Embedded

The structural patterns and variations within the records were algorithmically engineered to yield profound business answers when appropriately filtered and aggregate-modeled. The following 9 hidden initiatives are woven directly into the transaction layers:

- **Customer Segment Unit Economics:** Evaluating specific client cohorts against weighted historical lifetime parameters to flag the most net-profitable cohorts versus those consuming margins via high-cost logistics.
- **Product Category Portfolio Alignment:** Pinpointing low-volume, high-markup niches (such as Beauty and Skincare lines) versus high-revenue, narrow-margin anchors (such as Tier-1 Core Laptops and Mobile Hardware).
- **Systemic Return Drivers & Margin Leakage:** Mapping out why specific operational flows trigger product reversals—specifically highlighting size mismatches within Apparel, or fulfillment delays within isolated supply hubs.
- **Marketing Attribution Lifecycle Efficiency:** Quantifying customer capture costs against campaign spends to identify high-ROI organic/display configurations versus heavy discounting traps (e.g., specific margins eroded during seasonal flash promotions).
- **Logistical Bottlenecks & Service Levels:** Uncovering deep geographic variations where local infrastructure constraints or weather events create structural delivery delays, and tracing their negative feedback loops directly into customer retention indices.

3. Mathematical Definitions

To ensure logic validation across dashboards, the raw revenue mechanics adhere to strict core retail formulations:

$$Gross_Sales = Unit_Price \times Quantity$$

$$\text{Discount_Amount} = \text{Gross_Sales} \times \text{Discount_Percent}$$

$$\text{Net_Sales} = \text{Gross_Sales} - \text{Discount_Amount}$$

$$\text{Order_Profit} = \text{Net_Sales} - \text{Total_Product_Cost}$$

4. Comprehensive Field Schema Definitions

Field Technical Name	Type / Range	Business Meaning & Rules	Intentionally Introduced Anomalies
order_id	String ORD-XXXXXX	Unique alphanumeric transaction identifier generated at checking point. Primary indexing key.	None. Maintained strictly to trace duplicate records.
order_date	Date YYYY-MM-DD	The exact date the customer finalized transaction placement via the digital marketplace interface.	Invalid dates (e.g., 2024-02-30), mixed delimiters (slashes vs. hyphens), and system reset errors (0000-00-00).
customer_id	String CUST-XXXXX	Foreign relational key linking tracking attributes back to unique buyer identities.	Missing values (~6% Nulls) simulating session capture failure logs. Critical field break.
customer_segment	Categorical 6 Distinct	Proprietary buyer categorization based on historic behavioral tracking (e.g., Tech Enthusiasts).	Inconsistent text casing (mixed UPPERCASE, lowercase) and trailing/leading whitespace padding.
customer_acquisition_date	Date YYYY-MM-DD	The historic date the customer first registered an official profile within the OmniStore platform ecosystem.	Generally stable; can be checked against order dates for cohort structural verification.
customer_country	String Country Codes	Geographic country designation for shipping destinations and regional corporate cross-border tax assignment.	Inconsistent labeling codes. Blends full strings (United States) with abbreviations (US, UK, DE).
customer_state	String Text	The localized state, province, or primary internal geographic sub-district associated with delivery.	Inconsistent capitalization; occasional missing fields if country wide lookup failed.
customer_city	String Text	The municipal postal delivery entity name of the customer destination.	Typographical variations & abbreviations (e.g., 'Los Angeles' encoded as 'LA', 'Toronto' as 'Torontoo').

Field Technical Name	Type / Range	Business Meaning & Rules	Intentionally Introduced Anomalies
customer_zipcode	String Alphanumeric	Postal sorting code. Essential for routing matrix and localized structural delivery assignments.	High missingness (~12% Null values) representing customer omissions or uncaptured fields.
region	Categorical 5 Macro Regions	Broader territorial market zone grouping (e.g., East, West, Midwest, North, Europe, APAC).	Varying text inputs and lowercase mismatches across automated logs.
sku	String SKU-XXXX-XXX	Stock Keeping Unit. Universal commercial product standard cataloging identification string.	Generally stable, but links items with hidden cost-to-price parameters.
product_category	Categorical 4 Main Classes	Broad commercial categorization taxonomy (Electronics, Apparel, Home & Kitchen, Beauty & Personal Care).	Intentional system typos (e.g., 'Electrnics', 'Home/ Kitchen', and 'Apparel & Clothing'). Re-mapping required.
product_subcategory	Categorical Text	Granular category classification branch (e.g., Smartphones, Outerwear, Skincare, Cookware, Fitness).	Inconsistent case applications and rare text errors.
product_name	String Text	The localized product name visible to end-users on storefront detail product pages.	Clean string name references, but carries trailing white spaces occasionally.
quantity	Integer -99 to 5	The specific count of product units added inside the single row transactional bundle.	Invalid negative integer values (-99) introduced as clear database placeholder entry errors.
unit_price	Numeric / Mixed Float	The standard retail base cost charged per unit item before any regional line reductions or discounts.	Mixed data types. Rare records contain the raw string text flag "STUCK_VAL" .
unit_cost	Numeric Float	The underlying procurement or assembly production cost sustained by the company to build or acquire the item.	Fixed base numbers based on SKU catalog, providing ground-truth margin validation benchmarks.
gross_sales	Numeric Float	Total raw price accrued prior to promotional calculation adjustments. Should equal Price × Qty.	Extreme value outliers (15x scale multipliers) simulating automated point-of-sale terminal script errors.

Field Technical Name	Type / Range	Business Meaning & Rules	Intentionally Introduced Anomalies
discount_percent	Numeric 0.00 to 0.50	The fractional percentage discount deducted from the individual order item pricing row.	Varies significantly based on specific target marketing runs.
discount_amount	Numeric Float	The raw absolute financial discount amount calculated. Deducted from Gross Sales values.	Needs algorithmic math audit to check for internal logical alignment with discount rates.
net_sales	Numeric Float	The final actual top-line revenue collected by the retail group for the row transaction.	Missing values (~4% Null entries). Breaks financial math chains. Imputation or reconstruction required.
total_product_cost	Numeric Float	Total financial cost outflow required to fulfill product inventory. Cost × Qty.	Requires calculation audit against raw quantity parameters.
order_profit	Numeric Float	Calculated line-item net contribution margin. Can yield negative numbers indicating structural losses.	Extreme negative outliers injected matching structural point-of-sale calculation breakdowns.
marketing_campaign	String Camp IDs	Tracking identifier mapping the marketing initiative that successfully drove customer order attribution.	Contains 'NO_CAMP' for baseline organic traffic streams.
marketing_channel	Categorical Text	The channel medium carrying the creative marketing placement (Email, Paid Search, Social Media, Influencer, Display, Organic).	Inconsistent casing profiles across legacy channels.
marketing_cost_per_click	Numeric 0.00 to 0.60	The micro attribution cost paid out per digital click response for that channel type framework.	Consistent base variables for calculating marketing budget efficiency metrics.
shipping_mode	Categorical 4 Modes	The operational shipment courier speed framework requested (Standard, Express, Overnight, Economy).	Inconsistent case presentation values across database layers.
shipping_date	Date / Mixed YYYY-MM-DD	The day fulfillment centers packaged and handed off inventory parcels to third-party shipping couriers.	Missing strings / Blanks (~8%) representing missing scanning event logs.

Field Technical Name	Type / Range	Business Meaning & Rules	Intentionally Introduced Anomalies
delivery_date	Date YYYY-MM-DD	The verified timestamp marking safe customer-side receipt or home locker final dropping.	Subject to similar calendar format breaks and invalid date strings as order dates.
actual_shipping_days	Mixed / Text Mixed format	Calculated span from outbound courier dispatch to final doorstep drops. Key metric for logistics efficiency.	Mixed Unit formatting. Alternates between standard clean integers and string annotations like "5 days" .
shipping_cost	Numeric Float	The internal net delivery cost invoiced to OmniStore by regional logistic couriers.	Consistent float values based on logistical weights and distances.
shipping_charge_to_customer	Numeric Float	The delivery fee billed directly to the customer's payment profile (often zero due to loyalty tier incentives).	Consistent numerical matrix layout.
payment_method	Categorical 5 Variants	The financial mechanism leveraged to settle order transactions (Credit Card, PayPal, Apple Pay, Debit Card, Store Credit).	Text variation strings and case inconsistency issues.
customer_satisfaction_score	Integer / Ordinal 1 to 5	The post-delivery customer feedback score captured via email surveys. Scale ranges from 1 (Poor) to 5 (Excellent).	High structural missingness (~14% Nulls). Missing scores follow non-response behaviors.
order_returned	Binary Toggle 0 or 1	Boolean flag logging if the order was rejected and processed back into physical fulfillment warehouses.	Clean boolean integer values, serving as target vectors for returns risk modeling.

5. Duplicate Architecture Details

During database construction, an additional 120 rows were intentionally injected to stress test row-deduplication protocols. Analysts must differentiate between two distinct duplicate frameworks:

- Exact Duplicate Matches:** 60 records are precise line-for-line mirror images across all column variables, caused by web-app double-submission clicks. These must be purged cleanly via unique key constraints.
- Partial Duplicates / Audit Variances:** 60 records replicate order identities and item details but show subtle variations in `order_profit` parameters. These simulate localized terminal entry adjustments and require precise multi-conditional deduplication strategies.